Application of Curiosity Driven Exploration Algorithms for Hardware Interference Identification

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Outline

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Context

- The shift from single-core to multi-core architectures is essential in safety-critical embedded systems in multiple domains.
- However, this transition introduces new complexities due to tasks running in parallel competing for shared resources ; inter-core interference.
- The AMC 20-193 standard requires identifying all sources of interference that can occur when executing tasks in parallel and addressing them.

Purpose

The purpose of the study is to demonstrate the ability of curiosity-driven exploration algorithms to build a map from complex system behaviors to their causes.

- Great diversity of interference features involving shared combinations of shared resources by using available detailled information from the simulated platform.
- With limited experimental budget, more behaviors than sampling entries of the simulator randomly.
- Guidance for analyzing the results.

Interference sources

Definition

We call a "pair of programs" a couple of two programs of code S_1 and S_2 allocated respectively to the C_1 and C_2 cores of an execution platform. S_1 and S_2 are independent code programs. In particular, they do not exchange data.

One can distinguish distinct categories of interference occurring because of contention on shared resources :

- **Cache contention :** multiple cores compete for space in shared caches, evicting each other's data and causing cache misses.
- **Memory contention :** multiple cores access a shared memory simultaneously (such as the main memory), leading to a bottleneck due to limited memory bandwidth.
- **Bus contention :** cores compete for access to a shared bus, resulting in bandwidth limitations and delays in data transfer.
- **I/O Contention :** multiple processes try to perform I/O operations at the same time, leading to delays and reduced I/O performance.

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Our simplified simulator

Our simulator model contains the following **resources** :

- two cores,
- their respective private L1 caches,
- a shared L2 cache unit
- DDR
- DDR controller,
- an interconnect model.

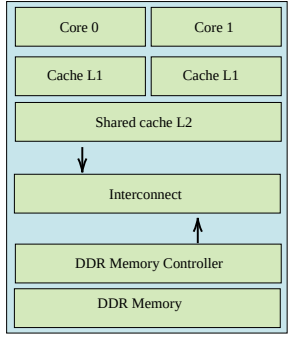


Figure – Simulated dual-core architecture used in this work.

How the simulator works

- Takes simplified assembly memory instructions
- One models distinct sets of addresses for the cores, i.e., core 0 : 0,9 and core 1 : 10,19.

```
1 program_core_0 = {23:('read', 11),37:('read',17),49:('write',6)}
2 program_core_1 = {12:('write', 24),18:('read',39),32:('write',37)}
```

- Outputs description of interference events for all resources.

Resources	Description	Features
L2 cache interference	Cross-core eviction tracking	set_idx, evicted_core_id, evicted_instr_id, evicted_addr, causing_core_id, causing_addr
BUS contention	Delayed requests due to bus bandwidth limitations	nb_competing_requests
DDR interference	Delayed requests due to bank conflicts	row,bank,addr,req_type,scheduled_delay,c ore_id
DDR scheduler interference	Delayed requests due to higher priority requests	bank,addr,req_type,scheduled_delay,core_id, core_attacker,type_attacker, addr_attacker,bank_attacker,row_attacker

Table – Events tracked by the simulator

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Overview of the method

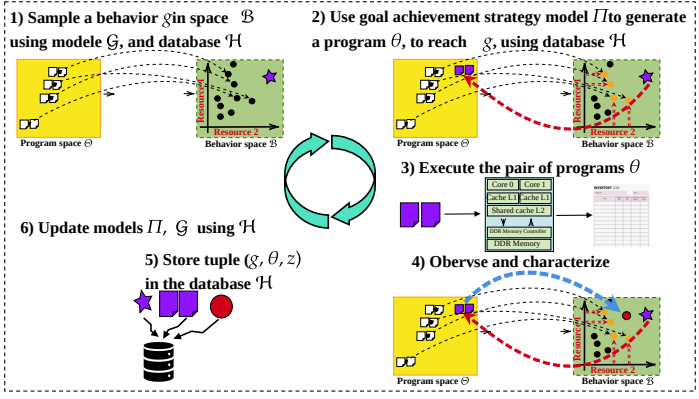


Figure – IMGEP for our case of interference identification, based on the general description of IMGEP

Behavior space and goal sampling

- We aim at covering the behavior space $\mathcal{B}$ of interference on the architecture.
- The behavior space $\mathcal{B}$ thus made from several parts $\mathcal{B}_r$ for each resource r : $\mathcal{B} = \cup_r^n \mathcal{B}_r$. For every experiment on the simulator, events for resources r are stored in part $\mathcal{H}[r]$ of $\mathcal{H}$.
- One wants to discover as many interference as possible among all the possible combinations of resources,
- One chooses to bias the discovery process by sampling goals.

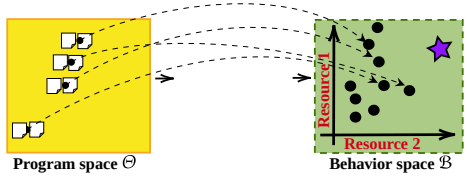


Figure – Goal sampling process. The purple star represents the goal in the behavior space. The axis indicate the agent tries to reach an experiment outcome with at least one interference for each required resource.

Goal sampling in two steps

One wants to found a pair of program that exhibits the chosen combination a^*.

- The powerset $\mathcal{P}(\mathcal{C})$ of the resource set $\mathcal{C}$ ={**DDR, DDR scheduler, Cache, Bus**} contains $2^4 - 1 = 15$.
- Example :A experience outcome qualified by a_4 means that at lest an interference on the L2 cache was found, but none for the other resources.
- The random variable g of support $\mathcal{S} = \cup_j^{card(\mathcal{P}(\mathcal{C}))}\{a_j^*\}$ determines what goal the agent chooses.
- Needs to find an optimal law of g $\forall a^* \in \mathcal{S}, \mathbb{P}_{\mathcal{H},g}(\{a^*\}) = \mathbb{P}_{\mathcal{H}}(g = a^*) \in [0, 1]$

Reference	resource combinations
a_1	**BUS**
a_2	**DDR**
a_3	**DDR scheduler**
a_4	**L2 cache interference**
a_5	**DDR, BUS**
a_6	**DDR scheduler BUS**
a_7	**L2 cache, BUS**
a_8	**DDR, DDR scheduler**
a_9	**L2 cache, ddr**
a_{10}	**L2 cache, DDR scheduler**
a_{11}	**DDR, DDR scheduler, BUS**
a_{12}	**L2 cache, DDR, BUS**
a_{13}	**L2 cache, DDR scheduler, BUS**
a_{14}	**L2 cache, DDR, DDR scheduler**
a_{15}	**L2 cache, DDR, DDR scheduler, BUS**

Table – resource combinations

Goal sampling in two steps

One wants to found a pair of program that exhibits the chosen combination a^*.

- Uniform sampling over $\mathcal{S}$:
  $\mathbb{P}_{\mathcal{H},g}(\{a^*\}) = \frac{1}{card(\mathcal{S})}, \forall a^* \in \mathcal{S}$
- Sampling based on exploration difficulty :
  $\mathbb{P}_{\mathcal{H},g}(\{a^*\}) = \frac{exp(-n_{\mathcal{H}}(a))}{\sum_j exp(-n_{\mathcal{H}}(a_j))}, \forall a^* \in \mathcal{S}$
- Diversity progress based sampling :$\mathbb{P}_{\mathcal{H},g}(\{a^*\}) = \frac{n_{\mathcal{H}}(a)}{\sum_j n_{\mathcal{H}}(a_j)}, \forall a^* \in \mathcal{S}$

Reference	resource combinations
a_1	BUS
a_2	DDR
a_3	DDR scheduler
a_4	L2 cache interference
a_5	DDR, BUS
a_6	DDR scheduler BUS
a_7	L2 cache, BUS
a_8	DDR, DDR scheduler
a_9	L2 cache, ddr
a_{10}	L2 cache, DDR scheduler
a_{11}	DDR, DDR scheduler, BUS
a_{12}	L2 cache, DDR, BUS
a_{13}	L2 cache, DDR scheduler, BUS
a_{14}	L2 cache, DDR, DDR scheduler
a_{15}	L2 cache, DDR, DDR scheduler, BUS

Table – resource combinations

Goal sampling in two steps

One when samples vectors for the each of the resource of the combination.

- Firstly, the agent samples a^* using the law $\mathbb{P}_{\mathcal{H},g}$ of g.
- Secondly, it samples vectors g_r associated to the resources or distinct type events $r \in a$.

$$g_r \sim \mathcal{U}([f_1 \cdot (min_{\mathcal{B}_r} z)_1, f_2 \cdot (min_{\mathcal{B}_r} z)_2]) \otimes \cdots \otimes \mathcal{U}([f_1 \cdot (min_{\mathcal{B}_r} z)_{\dim(\mathcal{B}_r)}, f_2 \cdot (max_{\mathcal{B}_r} z)_{\dim(\mathcal{B}_r)}])$$

Goal achievement strategy

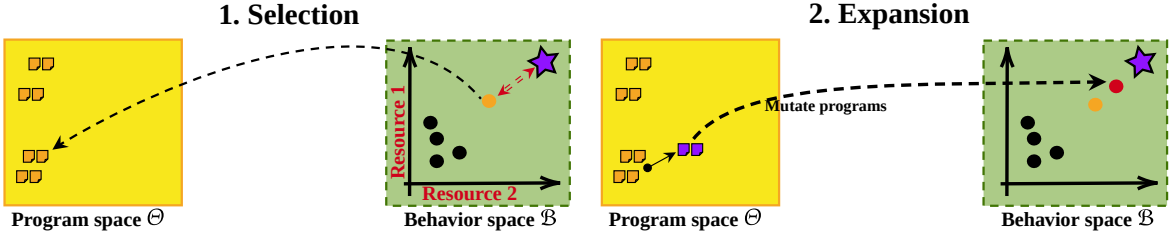


Figure – Goal achievement strategy for a combination of two resources. One selects the closest outcome from the database $\mathcal{H}$ (yellow), and one expands the history by producing a candidate parameter (purple)

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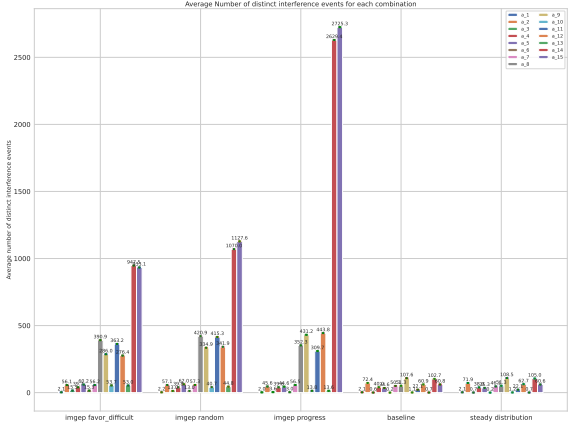
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Results

Number of distinct events for each resource combinations. This results from an example run of $N = 2 \cdot 10^4$ IMGEP iterations and a warming set of $N_{init} = 1000$ iterations of random program synthesis, and 3 random mutations from the mutation operator.



- IMGEP succeeds in finding multiple interference cases for every resource combination.
- As opposed to other steady sampling distributions.

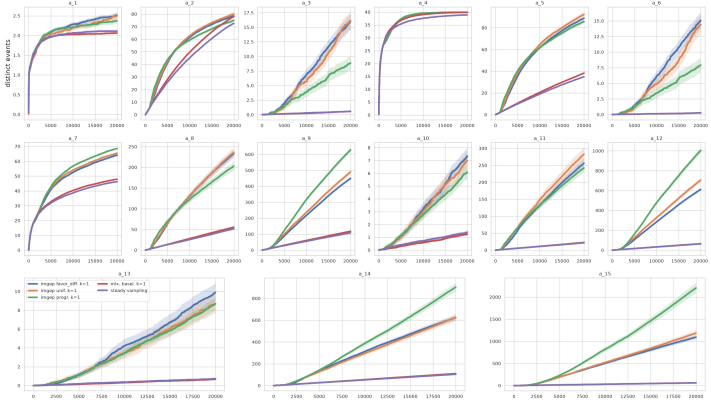


Figure – Evolution over the exploration of the number of distinct events for the different combinations of interference types. This results from the same example runs than presented on figure **??**

Analysis of the dataset

We demonstrate a prospective analysis method for assessing the nature of the collected intereference events.

- What are most encountered features for an interference combination $a \in \mathcal{S}$?
- $\implies$ Analyze Distribution of vectors $\{r : z_r \in \mathcal{B}(r) | \forall r \in a\}$.
- One can end up with vectors containing too many features to be easily analyzed. $\implies$ One keeps only the most relevant features before observing their distribution.

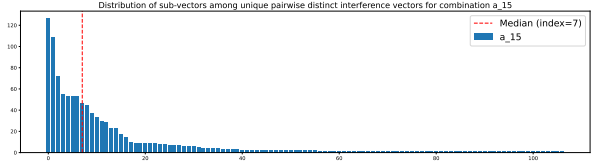


Figure – Among the set of pairwise distinct vectors for combination $a_{15} \in \mathcal{S}$, distribution of the subvectors featuring a selection of values. We see that median is 7, meaning that only 7 distinct vectors are responsible for 50 pourcents of the interference.

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	DDR :bank	DDR sched. :bank	DDR sched. :addr	DDR sched. :bank attacker	L2 cache :evicted core id	L2 cache :causing core id
1	5	5	13	5	1	0
2	2	2	2	2	1	0
3	5	5	13	5	0	1
4	4	4	4	4	1	0
5	1	1	9	1	1	0
6	5	5	5	5	1	0
7	5	5	5	5	0	1

Table – Vectors of features inducing the 50 percent quantile of the interference for combination a_{15}. Only a few features from table are considered, as one wants to perform a tracktable analysis.

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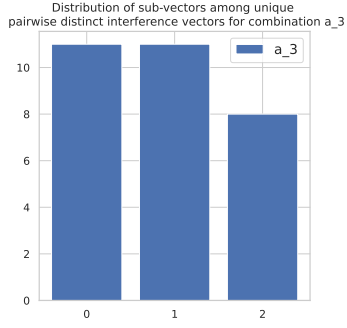


Figure – Among the set of pairwise distinct vectors for combination $a_3 \in \mathcal{S}$, distribution of the subvectors featuring a selection of values.

Analysis of the dataset

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bank	addr	bank attacker	row attacker
0	16	1	5
4	4	0	3
4	4	1	1

Limitations

- The goal achievement strategy also consists in mutating programs based on the goal.
- The system could be more realistic, and a transition on a high fidelity simulator like gem5 could benefit the audit of this project.

Timing data

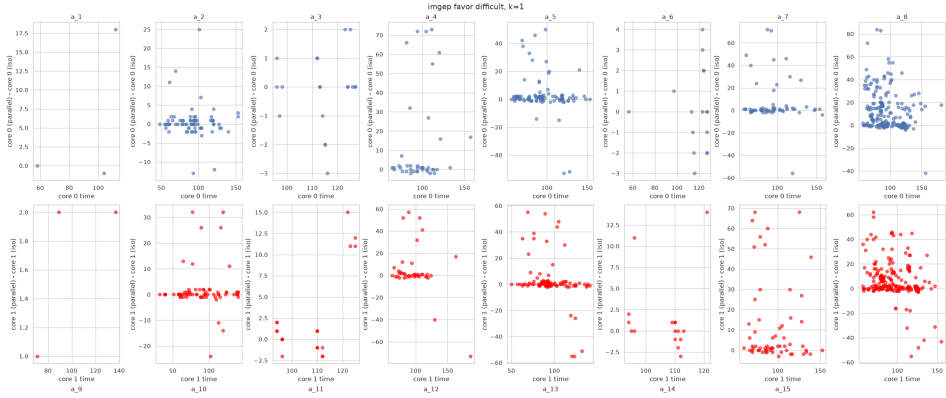


Figure – One observes that extending time does not determine the source of interference

Timing data

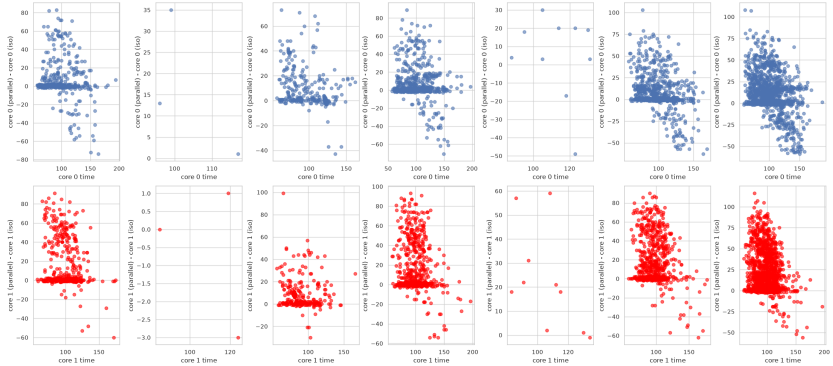


Figure – One observes that extending time does not determine the source of interference

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Conclusion

- The presented algorithm archiecture Intrinsically Goal Exploration Process relies on simple machine learning models to structure an autonomous agent that fixes his own goals, and on the establishment of a behavior and a parameter space
- Results demonstrate that automated discovery algorithms can successfully explore behaviors of a simplified simulated model of dual-core architecture, and encourage the idea of exploring more complex simulated platfoms reproducing real multicore architectures
- This approach encourages the idea synthesizing a tool to help researchers understand mechanisms that are sources of interference.